



SUPERCARS

USER MANUAL
GEN 3 SUPERCARS



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DEAR iRACING USER,

Congratulations on your purchase of the Gen 3 Supercar! From all of us at iRacing, we appreciate your support and your commitment to our product. We aim to deliver the ultimate sim racing experience, and we hope that you'll find plenty of excitement with us behind the wheel of your new car!

The following guide explains how to get the most out of your new car, from how to adjust its settings off of the track to what you'll see inside of the cockpit while driving. We hope that you'll find it useful in getting up to speed.

Thanks again for your purchase, and we'll see you on the track!



CHASSIS



**FRONT AND REAR DOUBLE A-ARM WITH
COILOVER SPRING AND ANTI-ROLL BAR**

LENGTH
4881 mm
192.2 in

WIDTH
1960 mm
77.2 in

WHEELBASE
2767 mm
109 in

DRY WEIGHT
1340 kg
2954 lbs

WET WEIGHT
WITH DRIVER
1439 kg
3173 lbs

POWER UNIT



NATURALLY ASPIRATED LS V8

DISPLACEMENT
5.7 Liters
347 cid

TORQUE
508.6 lb-ft
690 Nm

POWER
600 bhp
447 kW

RPM LIMIT
7500



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INTRODUCTION

The information found in this guide is intended to provide a deeper understanding of the chassis setup adjustments available in the garage, so that you may use the garage to tune the chassis setup to your preference.

Before diving into chassis adjustments, though, it is best to become familiar with the car and track. To that end, we have provided baseline setups for each track commonly raced by these cars.

To access the baseline setups, simply open the Garage, click iRacing Setups, and select the appropriate setup for your track of choice. If you are driving a track for which a dedicated baseline setup is not included, you may select a setup for a similar track to use as your

baseline. After you have selected an appropriate setup, get on track and focus on making smooth and consistent laps, identifying the proper racing line and experiencing tire wear and handling trends over a number of laps.

Once you are confident that you are nearing your driving potential with the included baseline setups, read on to begin tuning the car to your handling preferences.

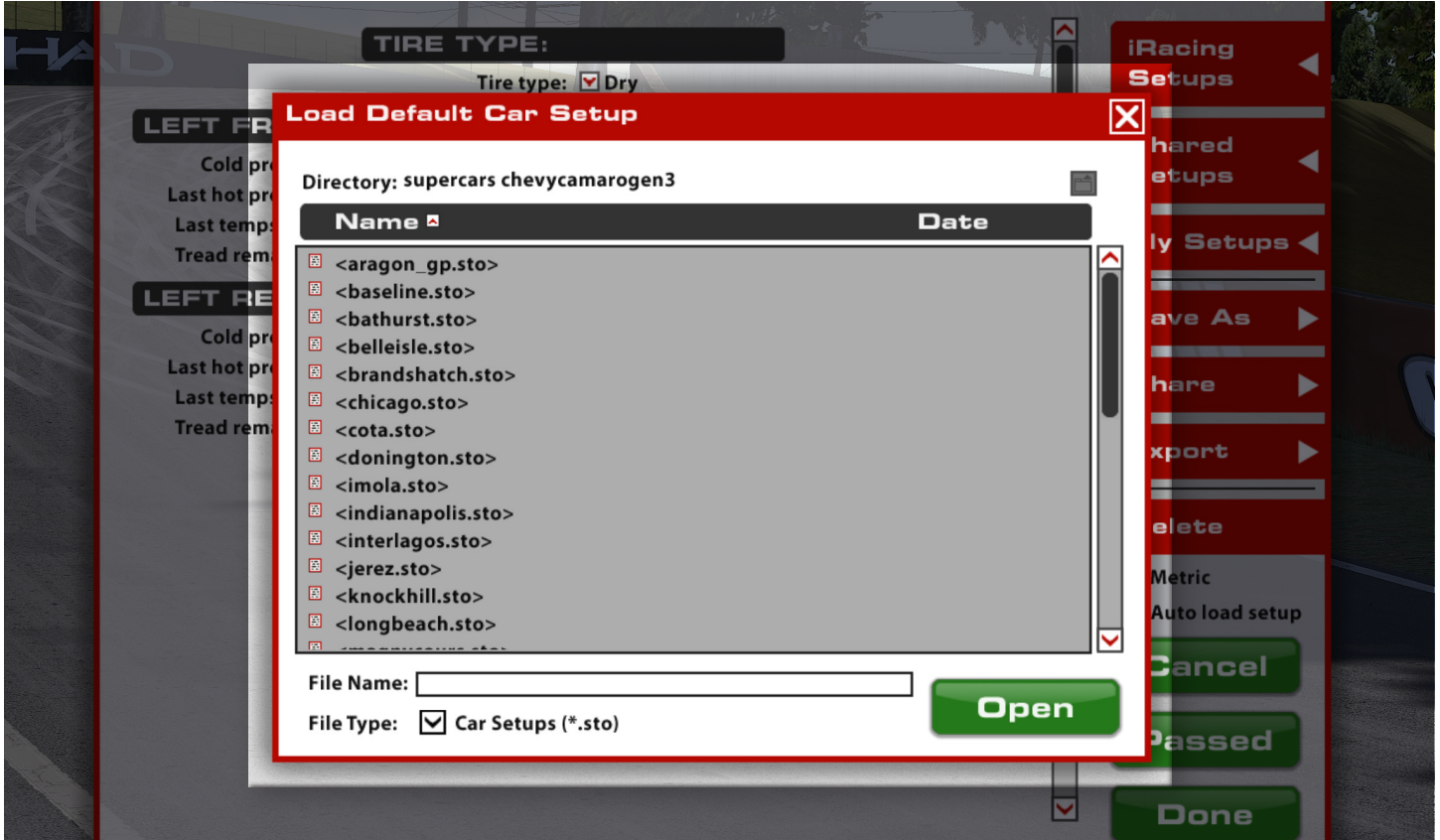
GETTING STARTED



Once you load into the car, press the clutch and select 1st gear. Give it a bit of throttle and ease off the clutch pedal to get underway. The Gen 3 Supercars use a clutchless 6-speed sequential transmission with reverse. Upshifts and downshifts don't require clutch or

throttle blip to move to the next gear, however in some situations it's beneficial to apply a slight throttle blip to downshifts to prevent the risk of wheel hopping. Upshifting is recommended when all shift lights illuminate in red, usually around 7500rpm.

LOADING AN iRACING SETUP



Upon loading into a session, the car will automatically load the iRacing Baseline setup <baseline.sto>. If you would prefer one of iRacing's pre-built setups that suit various conditions, you may load it by clicking Garage > iRacing Setups > and then selecting the setup to suit your needs.

If you would like to customize the setup, simply make the changes in the garage that you would like to update and click apply. If you would like to save your setup for future use click "Save As" on the right to name and save the changes. To access all of your personally saved setups, click "My Setups" on the right side of the garage.

If you would like to share a setup with another driver or everyone in a session, you can select "Share" on the right side of the garage to do so.

If a driver is trying to share a setup with you, you will find it under "Shared Setups" on the right side of the garage as well.

DASH PAGES

Both the Chevrolet Camaro and the Ford Mustang feature the same dash display in the same arrangement. Two pages are available, one to assist with race starts and one to provide information during on-track sessions.

RACE PAGE



Left Column	
Lap Time	The time for the previously-completed lap
Position/Lap Indicator	The gray box below the laptime displays the current session position in blue and the number of completed laps in white.
Fuel Lap	Amount of fuel used on the previous lap in Liters or Gallons
Engine Temp	Engine cooling water temperature in °C or °F
Center Column	
Laptime Delta Bar	Once a laptime has been recorded in a session the top of the dash will feature a colored bar originating at the center. If the current lap is faster than the reference the bar will be Green and move to the right. If the current lap is slower than the reference, the bar will turn red and move to the left of the screen.
Gain Loss +/-	The numerical lap time delta between the current lap and the reference lap is shown below the delta bar, colored green for a faster lap and red for a slower lap.
Pedal	The currently selected Throttle Pedal map. This mirrors the “PEDAL” setting in the F8 black box.
System Lights	These symbols illuminate when various systems are active. The only one that can be controlled in the sim is the blue Wiper indicator, shown just under the throttle pedal map setting. Enabling the pit limiter will flash the exterior light indicators.
Brake Bias	The current brake bias setting is shown at the bottom of the screen.
Right Column	
Ground Speed	Current speed in KPH or MPH
Gear	Currently selected transmission gear
Engine Oil	Engine oil temperature in °C or °F

LAUNCH PAGE

The Launch page is useful for standing starts, displaying helpful information to get a quick start at the beginning of a race.



Launch Page	
Pedal %	The position of the throttle pedal, shown as a percentage of the pedal's full travel
Engine Throttle %	How far the engine throttle is open, as a percentage of the throttle's full range. This may or may not be the same as the Pedal position.
Pedal	The currently selected Throttle Pedal map. This mirrors the "PEDAL" setting in the F8 black box.
Line Lock Status	Displays whether the Line Locker system is on or off. If enabled, the box will be purple and the current brake system pressure is shown.
Status Lights	These symbols illuminate when various systems are active. The only one that can be controlled in the sim is the blue Wiper indicator, shown just under the throttle pedal map setting. Enabling the pit limiter will flash the exterior light indicators.
Gear	Currently selected transmission gear
Brake Bias	The current brake bias setting is shown at the bottom of the screen.

SHIFT LIGHTS



Eight shift lights are situated above the digital display to indicate when the engine has reached the optimum RPM for shifting. Lights will illuminate from the outside towards the inside, with all eight lights flashing red when the engine has reached the shift point.

LOCK STATUS LIGHTS

During heavy braking, the shift LEDs will illuminate in various colors to represent wheel lockups. For all LEDs the lights will illuminate regardless of which wheel locks up on an axle (if only the LF locks, all “front” LEDs will illuminate):

- Red (Inner Two LEDs) - Rear wheel lockup
- White (Outer Two LEDs) - Minor Front wheel lockups
- Blue (Outer Two LEDs) - Severe Front wheel lockups



PIT LIMITER

When the pit limiter is active the display is similar to the Launch page but with minor changes.

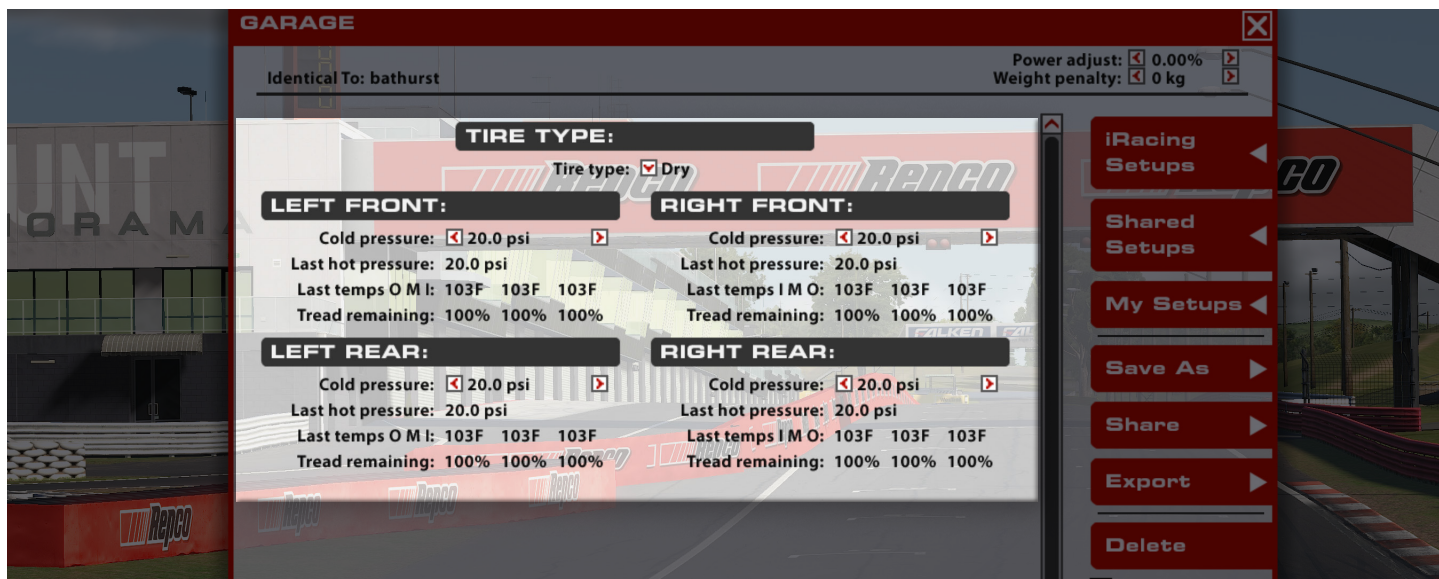


Pit Limiter	
Speed Lights	When the limiter is active the shift lights will change to an indicator of the vehicle's speed. If the car is traveling below the pit speed limit, the lights will be blue. If the car is at pit road speed limit, the lights will be green. If the car is speeding, the lights will be red.
Limiter Status Light	When the limiter is active a yellow LED will flash on the upper right of the dash module
Ground Speed	Vehicle speed, in kph or mph
Pedal	Current Throttle Map setting
Line Lock Status	Displays whether the line lock is on or off
Gear	Currently selected transmission gear
Throttle %	Current throttle pedal position, changes color based on position: Blue for 0-39%, Green for 40-60%, Red for 61-100%
Bias	Current Brake Bias Setting

ADVANCED SETUP OPTIONS

This section is aimed toward more advanced users who want to dive deeper into the different aspects of the vehicle's setup. Making adjustments to the following parameters is not required and can lead to significant changes in the way a vehicle handles. It is recommended that any adjustments are made in an incremental fashion and only singular variables are adjusted before testing changes.

TIRES



TIRE TYPE

Selects which type of tire is installed on the car when loaded into the world. Dry, or slick, tires are used for dry racing conditions while Wet tires are intended for raining and wet track conditions.

COLD PRESSURE / STARTING PRESSURE

The air pressure in the tires when the car is loaded into the world. Lower pressures will provide more grip but will produce more rolling drag and build temperature faster. Higher pressures will feel slightly more responsive and produce less rolling drag, but will result in less grip. Generally, higher pressures are preferred at tracks where speeds are higher while lower pressures work better at slower tracks where mechanical grip is important.

LAST HOT PRESSURE

When the car returns to the garage after an on-track stint, the tire pressure will be displayed as Hot Pressure. The difference between cold and hot pressure is a good way to see how tires are being loaded and worked while on track. Tires seeing more work will build more pressure, and paying attention to which tires are building more pressure and adjusting cold pressure to compensate can be crucial for optimizing tire performance.

LAST TEMPERATURES

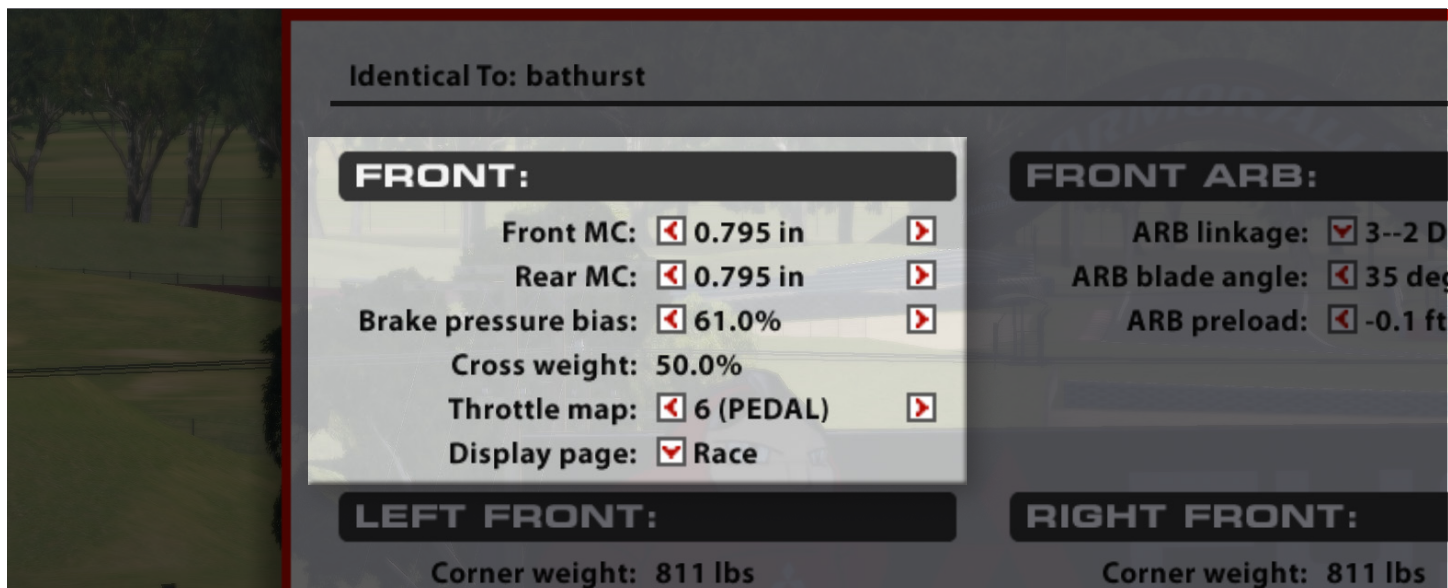
The tire carcass temperatures (measured within the tread) are displayed after the car returns from the track. These temperatures are an effective way to determine how much work or load a given tire is experiencing while on track. Differences between the inner and outer temperatures can be used to tune individual wheel alignment and the center temperatures can be compared to the outer temperatures to help tune tire pressure.

TREAD REMAINING

The amount of tread on the tire, displayed as a percentage of a new tire, is shown below the tire temperatures. These values are good for determining how far a set of tires can go before needing to be replaced, but don't necessarily indicate an under- or over-worked tire in the same way temperatures will.

CHASSIS

FRONT



FRONT MC

The Front Brake Master Cylinder size can be changed to alter the line pressure to the front brake calipers. A larger master cylinder will reduce the line pressure to the front brakes, which will shift the brake bias rearwards and increase the pedal effort required to lock the front wheels. A smaller master cylinder will increase brake line pressure to the front brakes, shifting brake bias forward and reducing required pedal effort to lock the front wheels. Adjustments to the Front Master Cylinder will automatically change the Brake Pressure Bias setting, so adjustments may be needed to reset the brake bias following a Master Cylinder change.

REAR MC

The Rear Brake Master Cylinder size can be changed to alter the line pressure to the rear brake calipers. A larger master cylinder will reduce the line pressure to the rear brakes, which will shift the brake bias forwards and increase the pedal effort required to lock the rear wheels. A smaller master cylinder will increase brake line pressure to the rear brakes, shifting brake bias rearward and reducing required pedal effort to lock the rear wheels. Adjustments to the Rear Master Cylinder will automatically change the Brake Pressure Bias setting, so adjustments may be needed to reset the brake bias following a Master Cylinder change.

BRAKE PRESSURE BIAS

Brake Bias is the percentage of braking force that is being sent to the front brakes. Values above 50% result in greater pressure in the front brake line relative to the rear brake line which will shift the brake balance forwards increasing the tendency to lock up the front tyres but potentially increasing overall stability in braking zones. This should be tuned for both driver preference and track conditions to get the optimum braking performance for a given situation.

THROTTLE MAP

The Throttle Map setting will alter how sensitive the throttle pedal feels at low input levels. Of the ten positions available setting #6 will result in a linear pedal behavior, with consistent engine torque delivery through the full range of throttle inputs. Settings 1-5 progressively get less sensitive in the lower throttle input range, with #1 being the least sensitive. Conversely, settings 7-10 become more sensitive, seeing more torque delivery earlier in the throttle pedal's travel.

DISPLAY PAGE

Changes the currently selected digital dash page. Two options are available as previously described in the dash configuration section of this manual.

FRONT ARB

Identical To: bathurst

Power adjust: ☐ 0.00%
Weight penalty: ☐ 0 kg

FRONT:

Front MC: ☐ 0.795 in ☐
Rear MC: ☐ 0.795 in ☐
Brake pressure bias: ☐ 61.0% ☐
Cross weight: 50.0%
Throttle map: ☐ 6 (PEDAL) ☐
Display page: ☒ Race

FRONT ARB:

ARB linkage: ☒ 3--2 Digress soft
ARB blade angle: ☐ 35 deg ☐
ARB preload: ☐ -0.1 ft-lbs ☐

LEFT FRONT:

Corner weight: 811 lbs
Ride height: 5.38 in

RIGHT FRONT:

Corner weight: 811 lbs
Ride height: 5.38 in

iRacing
Setups

Shared
Setups

My Setup

Save As

ARB LINKAGE

The front ARB Linkage can be set to one of six options as well as operating in two different ways: Linear and Digressive.

The first, more conventional, operation is the Linear mode using settings 2-1 (Linear Stiff) and 2-2 (Linear Soft). In these settings, the ARB will produce the same consistent rate in all conditions. As usual with ARB settings, the stiffer option will induce understeer while cornering while the softer option will reduce understeer.

The Digressive modes will alter the ARB stiffness based on the overall front ride height. When the front end is low, such as under heavy braking or driving through a dip, the bar will soften, reducing understeer at corner entry. As the front end rises, the bar will begin to stiffen. This can be very helpful when good turn-in is necessary without hurting corner-exit traction, but can also lead to handling instabilities if the front end moves too much through a corner, such as through a banked turn or over crests and dips. Four options are available for the Digressive mode, with 1-1 being the stiffest option and 3-2 being the softest. Setting 2-1 and 3-1 are mid-range settings, with 2-1 being the stiffer of these two options. Changing the setting will keep the same digressive rate curve, but stiffen or soften the bar overall.

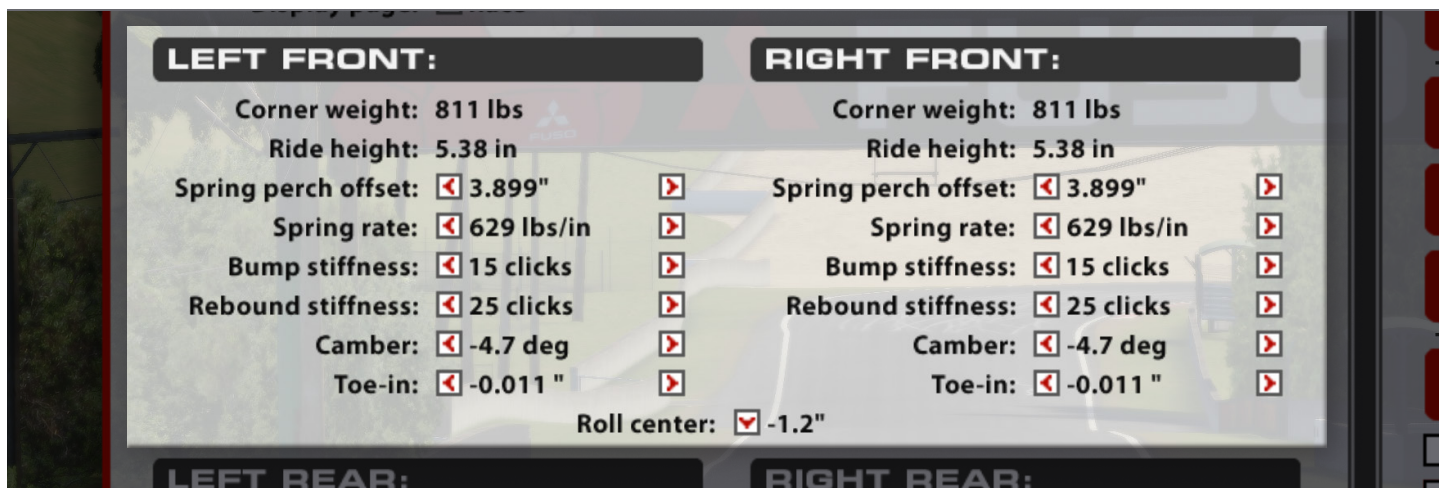
ARB BLADE ANGLE

The ARB Blades (or arms) can be changed to further tune the suspension roll stiffness beyond only the ARB size setting. This option changes the angle of the ARB arms relative to the ground with 0 Degrees being the softest and 90 degrees being the stiffest. Stiffer settings will result in a stiffer overall ARB system and more understeer, while softer settings will reduce understeer.

ARB PRELOAD

The ARB Preload is the static load in the bar while the vehicle is in the garage. Following chassis adjustments, this adjustment can be used to remove the static load in the bar to prevent asymmetrical or unexpected handling characteristics. Keep the value at or near zero for the most consistent performance.

FRONT CORNERS



CORNER WEIGHT

The weight underneath each tire under static conditions in the garage. Correct weight arrangement around the car is crucial for optimizing a car for a given track and conditions. Individual wheel weight adjustments and crossweight adjustments are made via the spring perch offset adjustments at each corner.

RIDE HEIGHT

Distance from the ground to the bottom surface of the car's floor on the front wheel centerline. Adjusting Ride Heights is key for optimum performance, as they can directly influence the vehicle's aerodynamic performance as well as mechanical grip. Increasing front ride height will allow for more weight transfer across the front axle, slightly increasing oversteer when cornering, while reducing front ride height reduces the weight transfer across the front axle, which can produce a more stable-feeling front end. Generally, keeping the front ride heights low, but still high enough to clear bumps and kerbs, is best.

SPRING PERCH OFFSET

Used to adjust the ride height at the corner of the car by changing the installed position of the spring. Increasing the spring perch offset will result in lowering the corner of the car while reducing the spring perch offset will raise the corner of the car. These changes should be kept symmetrical across the axle (left to right) to ensure the same corner ride heights and no change in cross weight. The spring perch offsets can also be used in diagonal pairs (LF to RR and RF to LR) to change the static cross weight in the car.

SPRING RATE

This setting determines the installed corner spring stiffness. Stiffer springs will result in a smaller variance in ride height between high and low load cases, but can hurt mechanical grip if the difference in max and minimum load is too great or too sudden. Typically the drawbacks of stiffer springs will become more pronounced on rougher tracks and softer springs in these situations will result in increased overall performance. Smoother tracks can sometimes see better performance with stiffer springs due to a more consistent chassis attitude and more responsive handling.

BUMP STIFFNESS

The bump stiffness setting is a paired adjustment controlling both the low and high speed compression damping characteristics of the damper. For the Supercars, setting "40" is minimum damping (least resistance to compression) while "0" is maximum damping (most resistance to compression). Increasing the bump stiffness will result in faster weight transfer to the wheel during transient movements, such as braking and direction change, with increased damping usually providing an increase in turn-in response but a reduction in overall grip in the context of front dampers. High speed compression damping will increase proportionally to the increase in low speed compression damping which will also result in harsher response to kerb strikes. At smoother tracks more bump stiffness (lower number) will typically increase performance while at rougher tracks or ones with aggressive kerbs less compression damping (higher number) can result in an increase in mechanical grip at the expense of platform control.

REBOUND STIFFNESS

The Rebound Stiffness setting is a paired adjustment to both low and high speed rebound damping characteristics. Increasing rebound damping will slow down the rate at which the damper extends in both low and high speed situations. A typical low damper speed situation would be as the car rolls back to level on a corner exit while a high speed situation would be when the suspension is extending after large kerb contact. Setting "40" is minimum damping (least resistance to extension) while "0" is maximum damping (most resistance to extension). While high rebound stiffness will result in improved platform control and overall chassis response it is important to avoid situations where the shock is too slow in rebounding as this will result in the tire losing complete contact with the track surface which can induce or exacerbate severe oscillations. The lack of aerodynamic downforce on the Supercars means it will usually perform better with lower rebound forces (higher click numbers). High speed cornering but will result in a loss of braking performance and necessitate a rearward shift in brake bias to compensate.

CAMBER

Camber is the vertical angle of the wheel relative to the center of the chassis. Negative camber is when the top of the wheel is closer to the chassis centerline than the bottom of the wheel, positive camber is when the top of the tire is farther out than the bottom. Due to suspension geometry and corner loads, negative camber is desired on all four wheels. Higher negative camber values will increase the cornering force generated by the tire, but will reduce the amount of longitudinal grip the tire will have under braking. Excessive camber values can produce very high cornering forces but will also significantly reduce tire life, so it is important to find a balance between life and performance. Increasing front camber values will typically result in increased front axle grip during mid to high speed cornering but will result in a loss of braking performance and necessitate a rearward shift in brake bias to compensate.

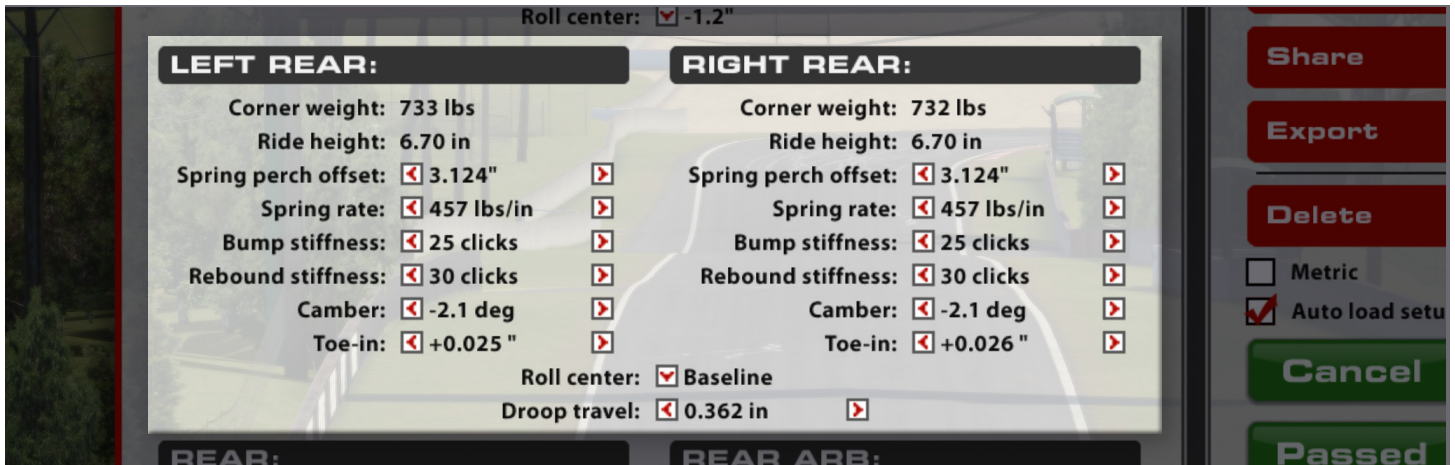
TOE-IN

Toe is the angle of the wheel, when viewed from above, relative to the centerline of the chassis. Toe-in is when the front of the wheel is closer to the centerline than the rear of the wheel, and Toe-out is the opposite. On the front end, adding toe-out (negative value) will increase slip in the inside tire and decrease straight-line stability while adding toe-in will reduce the slip and increase straight-line stability.

ROLL CENTER

The front Roll Center is a suspension geometry parameter that affects the amount of weight transfer during cornering. This can be run at the Baseline value, raised 40mm (1.6"), or lowered 30mm (1.2"). Raising the front Roll Center will increase front roll stiffness and induce understeer while lowering the front roll center will reduce front roll stiffness and induce oversteer.

REAR CORNERS



CORNER WEIGHT

The weight underneath each tire under static conditions in the garage. Correct weight arrangement around the car is crucial for optimizing a car for a given track and conditions. Individual wheel weight adjustments and crossweight adjustments are made via the spring perch offset adjustments at each corner.

RIDE HEIGHT

Distance from the ground to the bottom surface of the car's floor on the rear wheel centerline. Adjusting Ride Heights is key for optimum performance, as they can directly influence the vehicle's aerodynamic performance as well as mechanical grip. Increasing rear ride height will allow for more weight transfer across the rear axle, slightly increasing oversteer when cornering, while reducing rear ride height reduces the weight transfer across the rear axle, which can produce a more stable-feeling rear end, especially under heavy throttle application.

SPRING PERCH OFFSET

Used to adjust the ride height at the corner of the car by changing the installed position of the spring. Increasing the spring perch offset will result in lowering the corner of the car while reducing the spring perch offset will raise the corner of the car. These changes should be kept symmetrical across the axle (left to right) to ensure the same corner ride heights and no change in cross weight. The spring perch offsets can also be used in diagonal pairs (LF to RR and RF to LR) to change the static cross weight in the car.

SPRING RATE

Changes the installed coil spring rate. Without aerodynamic considerations, the rear springs will influence how much grip is available on throttle application and how the car reacts at turn-in. Stiffer springs will be more responsive at corner entry, shifting the balance towards oversteer but could hurt forward grip on throttle application, especially on low-grip or bumpy surfaces. Softer springs will give a more sluggish, understeery turn-in response, but will allow for more aggressive throttle application on corner exit.

BUMP STIFFNESS

The bump stiffness setting is a paired adjustment controlling both the low and high speed compression damping characteristics of the damper with identical ranges to those of the front dampers. Increasing the compression damping will result in faster weight transfer to the wheel during transient movements, such as accelerating and direction change, with increased damping usually providing an increase in response but a reduction in overall grip especially at corner exit traction in the context of rear dampers. Excessively stiff compression damping can cause very poor traction on rough tracks as it can result in large tire load variation and a reduction in overall grip. As with the front shocks, setting "40" is minimum compression damping and "0" is maximum compression damping.

REBOUND STIFFNESS

The rebound stiffness setting is a paired adjustment controlling both the low and high speed damping characteristics of the damper with identical ranges to those of the front dampers. Increasing rebound damping will slow down the rate at which the damper extends in both low and high speed situations. High rebound stiffness will result in improved platform control for aerodynamic performance and overall chassis response but it is important to avoid situations where the shock is too slow in rebounding as this will result in the tire losing complete contact with the track surface. This can be particularly detrimental during braking events and during the initial turn-in phase though an increase in rebound stiffness can help to "slow down" the change in pitch of the car as the brakes are applied, potentially increasing braking stability. As with the front shocks, setting "40" is minimum rebound damping and "0" is maximum rebound damping.

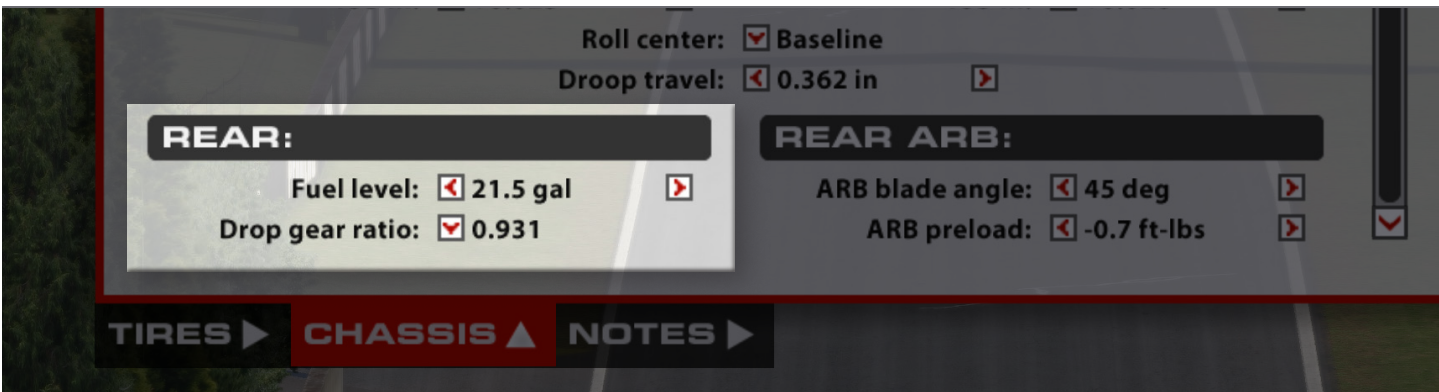
CAMBER

Rear Camber can be adjusted to alter the amount of lateral and longitudinal force the tire is capable of producing while cornering and accelerating. Running more negative camber will increase lateral force from the tire, helping to keep the car stable when cornering. This can hurt acceleration, however, as more negative camber will reduce the amount of traction the tire is capable of producing. Less negative camber will provide more traction on acceleration, but could result in the rear of the car feeling light through medium- to high-speed corners.

TOE-IN

Toe is the angle of the wheel, when viewed from above, relative to the centerline of the chassis. Toe-in is when the front of the wheel is closer to the centerline than the rear of the wheel, and Toe-out is the opposite. At the rear of the car it is typical to run toe-in. Increases in toe-in will result in improved straight line stability and a reduction in response during direction changes. Large values of toe-in should be avoided if possible as this will increase rolling drag and reduce straight line speeds. Generally, it is advised to keep the left and right toe values equal to prevent crabbing or asymmetric handling behavior.

REAR



FUEL LEVEL

The amount of fuel in the fuel tank when the car is loaded into the world.

DROOP TRAVEL

Suspension Droop is how far the suspension can extend when unloaded. By changing the amount of droop available, the inside rear wheel can be intentionally unloaded when cornering to help the car rotate into and through a corner. Increasing the Droop Travel setting will allow the suspension to extend further and keep the wheel loaded while cornering, decreasing the setting will restrict suspension movement and lift the inside rear wheel when cornering.

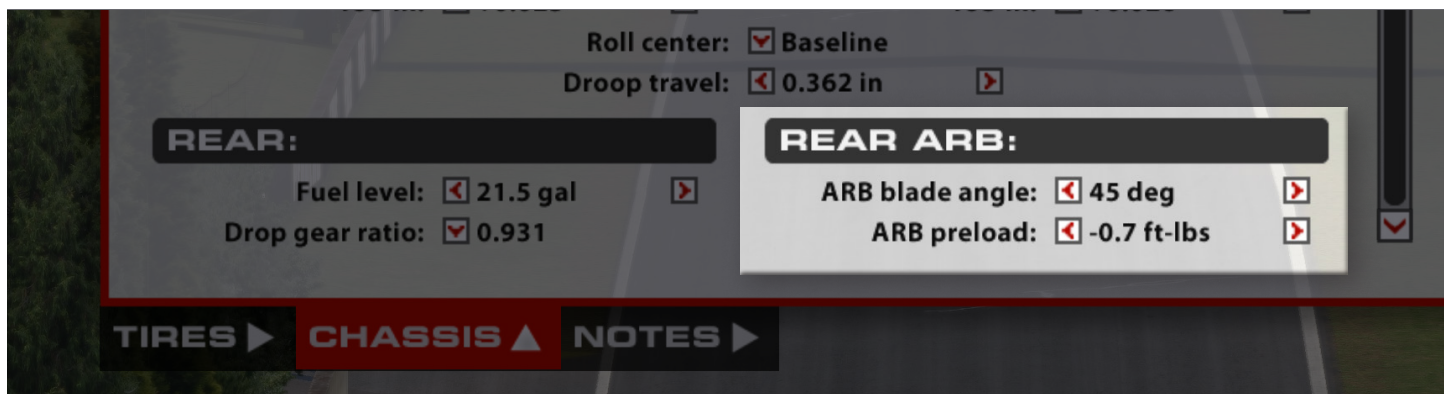
ROLL CENTER

The Rear Roll Center is a parameter of the rear suspension geometry that directly affects weight transfer and roll stiffness in the rear suspension. Higher Roll Center values will increase rear roll stiffness, reduce rear body roll, and increase oversteer when cornering, while lower Roll Center values will reduce roll stiffness, increase body roll, and increase understeer. The Roll Center setting is a very rough adjustment and may require an adjustment to rear ARB size to avoid compounding changes: Lowering the roll center may require a stiffer rear arb and raising the roll center may require softening the rear arb.

DROP GEAR RATIO

The Drop Gear is a set of quick-change gears that alter the final drive gear ratio. Higher ratios will provide better acceleration but a lower top speed, while lower ratios will increase top speed at the expense of acceleration. See the Setup Tips section for a list of gear recommendations from the real-world Supercars Series.

REAR ARB



ARB BLADE ANGLE

The ARB Blades (or arms) can be changed to further tune the suspension roll stiffness beyond only the ARB size setting. This option changes the angle of the ARB arms relative to the ground with 0 Degrees being the softest and 90 degrees being the stiffest. Stiffer settings will result in a stiffer overall ARB system and more oversteer, while softer settings will reduce oversteer.

ARB PRELOAD

The ARB Preload is the static load in the bar while the vehicle is in the garage. Following chassis adjustments, this adjustment can be used to remove the static load in the bar to prevent asymmetrical or unexpected handling characteristics. Keep the value at or near zero for the most consistent performance.

SETUP TIPS

This section is aimed toward helping users who want to dive deeper into the different aspects of the vehicle's setup.

DRIVING & SETUP TIPS

The Gen 3 Supercars are a large step away from the principles used in previous years for the series, with a very large drop in aerodynamic performance and more emphasis placed on mechanical grip and driver skill. When compared with the Gen 2 Supercars, aerodynamic loads can be halved, or even less, in similar conditions. Most corners will see so little aerodynamic performance that it can largely be ignored, changing how the car will be driven and setup from years past.

With such low aerodynamic performance, one of the biggest changes to the driver is the responsiveness and feel, especially at high speeds. Despite low aerodynamic loading, there is still a small amount situated near the rear of the car, which lends itself to a more understeery turn-in at high- and medium-speed corners compared to previous generations.

SETUP CONSIDERATIONS

Setup choices will be almost entirely based on maximizing mechanical grip and managing wheel loads. Softer springs and higher ride heights will result in more mechanical grip, with the Droop Limiter adjustment controlling how loaded the inside rear tire is while cornering.

Springs will see good performance when run as soft as the driver can handle, but stiff enough to keep the car from bottoming out under heavy loads. Rear springs can be used to tune in rear grip in both lateral and longitudinal grip, but at higher grip tracks can be run stiffer to help the car rotate into higher-speed corners.

Shocks can be helpful for controlling how fast the chassis pitches and rolls, but it's very important to remember the numbering convention for the adjustments: Lower "click" values are more damping forces. For example, if the front end dives too quickly then reducing the click value on front shock compression and rear rebound will slow the chassis pitch movement.

DROOP TRAVEL LIMITER

The rear Droop Travel setting is extremely effective at tuning to various grip levels and, for many tracks, can be the biggest adjustment for dialing a car in. Smaller values (more travel limit) will unload the inside rear wheel sooner at turn-in and induce more oversteer through the corner until steering input is relaxed for corner exit. Larger values will let the suspension extend further, keeping the inner rear wheel on the track and loaded. Generally, lower-grip tracks (like Street Circuits) will see better performance from running a higher Travel value and keeping the inner wheel loaded. Higher grip tracks where there is enough to keep the car stable on three wheels can see better performance from reducing the travel and lifting the inside rear wheel.

This adjustment can manifest a few different ways due to what it affects as the car is cornering. Most obvious is the unloaded inner rear wheel, dramatically shifting the dynamic crossweight to load the inner front and outer rear wheels, helping the car turn into the corner. Since the car has a spool instead of a differential, lifting the inner rear wheel also eliminates the "scrub" from this inside wheel since it will always rotate at the same speed as the outside rear wheel, causing understeer. It's important to test various settings at all types of tracks to see what works for the driver and how it affects the chassis feel.

DIGRESSIVE ARB

One of the more interesting adjustments on the Gen 3 Supercars is the variable-rate ARB mechanism in the front suspension. When in use, the ARB will have a softer rate when the front end is running lower, such as under heavy braking, and a stiffer rate when the front end rises up, such as on throttle. This is an excellent tool to use to overcome the car's inherent understeer at high-speeds, and when paired with good trail-braking technique can essentially eliminate the understeer entirely.

Despite the benefits, this does have a few drawbacks that become apparent at tracks with large dips and banked corners. If a turn happens to go through a dip that compresses the front end (or a crest that causes the nose to "pop" up), the ARB can change rate mid-corner, shifting the handling balance very suddenly. Similarly, if a corner is banked it will try to compress the front end, reducing the suspension's roll stiffness and causing the outer front corner to contact the racing surface. If this happens, it's important to consider the possible solutions to correct the issue. Stiffer springs will lift the front end in the corner, but could hurt mechanical grip in other corners, for example. Instead, a more effective fix would be to switch to a stiffer digressive option, increasing the bar rate throughout the range of travel and keeping the front roll stiffness higher.

DROP GEAR SETTINGS

The Gen 3 Supercars allow for 6 drop gear options to tune acceleration and top speed for any track. The real series mandates specific gears for each track, some of which (and their backup options) are listed below as starting points if needed:

Bathurst (Mount Panorama Circuit): 0.931, 0.909, 1.000
 Phillip Island: 1.000, 0.931, 1.042
 Sandown: 1.000, 0.931, 1.042
 Winton: 1.130, 1.074

